

RESEARCH ARTICLE

FDEM-TH3D: A three-dimensional coupled hydrothermal model for fractured rock

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Funding information

National Natural Science Foundation of China, Grant/Award Numbers: 11872340, 11602006 and 41731284; Beijing Natural Science Foundation, Grant/Award Number: 1174012; Chaoyang District Postdoctoral Science Foundation, Grant/Award Number: 2016ZZ-01-08; China University of Geosciences (Wuhan), Grant/Award Number: CUG170657

Summary

This paper proposes a three-dimensional coupled hydrothermal model for fractured rock based on the finite-discrete element method to simulate fluid flow and heat transport. The 3D coupled hydrothermal model is composed of three main parts: a heat conduction model for the rock matrix, a heat transfer model for the fluid in the fractures (including heat conduction and heat convection), and a heat exchange model between the rock matrix and the fluid in the fractures. Four examples with analytical solutions are provided to verify the model. A heat exchange experiment of circulating water in a cylindrical granite sample with one fracture is simulated. The simulation results agree well with the experimental results. The effects of the fracture aperture, fluid viscosity, and pressure difference on the heat exchange between the fluid and rock are studied. Finally, an application concerned with heat transport and fluid flow in fractured rock is presented. The simulation results indicate that the 3D fully coupled hydrothermal model can capture the fluid flow and temperature evolution of rocks and fluids.

KEYWORDS

3D finite discrete element method, fractured rock, heat exchange between fluid and rock, heat transfer, hydrothermal coupling, seepage

1 | INTRODUCTION

Many applications in environmental and energy engineering involve heat flow and transport due to fluid flow through fractured rock.^{1–9} Heat transport in fractured rock is strongly influenced by fluid flow, and the change in temperature of the fluid also affects the fluid flow in the fracture. The mutual influence between the heat transport and fluid flow is called hydrothermal coupling. Fluid flow in fractures affects the heat transfer in fractured rock through heat convection and ultimately affects the temperature distribution, and the temperature alters the fluid viscosity and thus affects the fluid flow in the fracture.

Due to the importance of hydrothermal coupling in fractured rock, several coupled models or algorithms have been proposed. For example, Jiang et al¹⁰ developed a novel three-dimensional transient model for studying the subsurface heat exchange process in EGS. Chen and Jiang¹¹ proposed a nonequilibrium thermal model for heat transfer in a porous subsurface-heated reservoir, in which the EGS reservoir is considered as an equivalent porous medium. Cui et al¹² performed a numerical analysis of coupled hydrothermal problems in geotechnical engineering. However, these models are mainly based on a continuous medium, which cannot consider the effects of fractures on fluid flow and heat transport, especially in rock with sparse fractures or large faults.

Therefore, other models based on a discontinuous medium or a dual porosity medium have been proposed to perform coupled hydrothermal analyses. For example, Xu et al.¹³ proposed a simplified coupled hydrothermal model for enhanced geothermal systems. Tomac and Gutierrez² presented a coupled convective-conductive heat flow and transport model using the discrete element method (DEM). Luo et al.¹⁴ studied the effect of fracture surface roughness on fluid flow and heat transfer processes in fractured rocks using a discrete fracture network model. Heinze and Hamidi¹⁵ studied fluid flow and heat transport in fractured porous media using a dual porosity model and a local thermal non-equilibrium model. These models mainly focus on coupled hydrothermal analysis and do not consider crack initiation and propagation or the interaction of complex fractures when they are extended to the coupled thermohydromechanical (THM) case.

Munjiza^{16,17} proposed an approach for simulating solid fracturing, the combined finite-discrete element method (FDEM), which can simulate crack initiation and propagation through joint element breakage. The method combines the advantages of the finite element method for modeling solid deformation and the discrete element method for processing contact interaction between blocks. In recent years, the FDEM has been widely used in the field of rock mechanics, especially in relation to the fracturing of rock.¹⁸⁻²⁴ However, the FDEM can only be used to perform pure fracture mechanics calculations. Several coupled hydromechanical or thermal-mechanical models have been developed for the FDEM²⁵⁻³¹ to simulate solid deformation and fracturing driven by a fluid or temperature. However, the FDEM still cannot be used to address 3D coupled hydrothermal problems.³² In this paper, we present a three-dimensional coupled hydrothermal model (FDEM-TH3D) for simulating heat and fluid transport in fractured rock. A complete approach for simulating rock fracturing under the effect of hydrothermal-mechanical coupling can be constructed by combining the 3D coupled hydrothermal model with the FDEM. Because rock deformation and fracturing can be directly simulated by the FDEM, we focus on the 3D coupled hydrothermal model; the coupled THM case is not discussed in this paper.

This paper is organized as follows. First, the heat transfer model is introduced, which includes four parts: a heat conduction model in the rock matrix, a heat conduction model for the fluid in the fracture, a heat convection model for the fluid in the fracture, and the heat exchange between the fluid and the rock. The calculation procedure of the 3D coupled hydrothermal model is then presented. Subsequently, four examples with analytical solutions are presented to verify the 3D coupled hydrothermal model for fractured rock. An experiment of heat exchange between fluid and rock is simulated, and the simulation results are compared with the experimental results. The influences of the fracture aperture and the fluid viscosity on the heat transfer between the fluid and the rock are investigated. Finally, an application of heat transport and fluid flow in rock containing a fracture network is presented.

2 | HEAT TRANSFER MODEL

According to Fourier's law of heat conduction, the heat flow rate per unit area in the direction i can be expressed as follows:

$$q_i = -k_{ij} \frac{\partial T}{\partial x_j}, \quad (1)$$

where q_i is the heat flow rate in the i direction, k_{ij} is the thermal conductivity tensor, and T is the temperature.

For any given mass M , the temperature change can be written as follows:

$$\frac{dT}{dt} = \frac{Q_{total}}{CM}, \quad (2)$$

where Q_{total} is the total nodal heat into the mass M per unit time, C is the specific heat, and M is the mass.

2.1 | Heat conduction in the rock matrix

In the three-dimensional FDEM, the continuum is discretized into many tetrahedral elements, and joint elements are inserted between the neighboring tetrahedral elements (Figure 1). A heat conduction calculation model is constructed based on the topological connections between the tetrahedral elements and the joint elements shown in Figure 1. The temperature in a tetrahedral element is expressed by the temperatures of the four nodes that form it. The temperature in the continuum is represented by the temperature of the discrete nodes, as shown in nodes 1 to 7 in Figure 1.

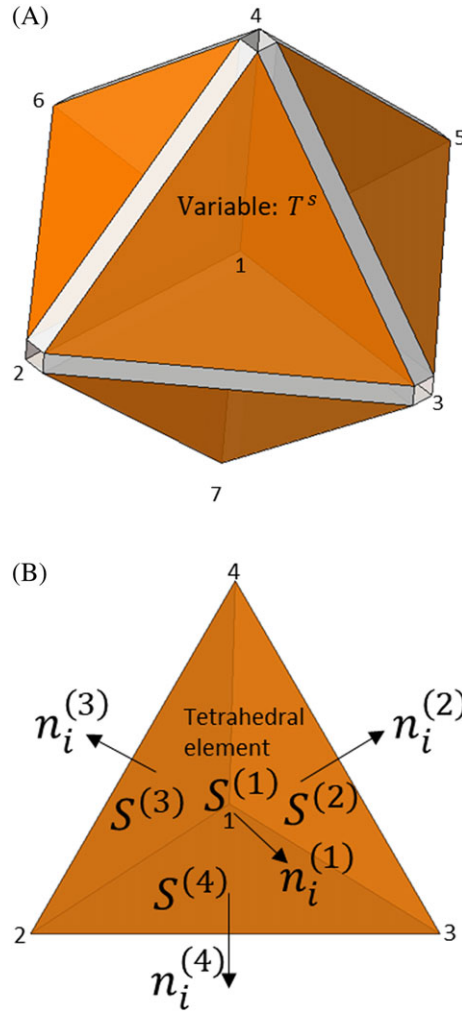


FIGURE 1 Topological connection of tetrahedral elements and joint elements in the FDEM-TH3D model for heat conduction in the rock matrix. A, Fracture surface in a rock. B, Mesh of the fracture surface [Colour figure can be viewed at wileyonlinelibrary.com]

Taking node 1 in Figure 1 as an example, the tetrahedral elements T1234, T1354, T1564, T1624, T1237, T1357, T1567, and T1627 are connected to node 1. Because the temperatures of nodes 2, 3, 4, 5, 6, and 7 may be different from that of node 1, heat conduction may occur in these tetrahedral elements. Taking the tetrahedral element T1234 as an example and assuming that the temperature field in the element has a linear distribution, the temperature gradient in the tetrahedral element is constant and is given by

$$\frac{\partial T^s}{\partial x_i} = \frac{1}{V_s} \int \frac{\partial T^s}{\partial x_i} dV. \quad (3)$$

T^s is the temperature of the rock matrix and V_s is the volume of the tetrahedral element. According to the divergence theorem, Equation 3 can be written as

$$\frac{\partial T^s}{\partial x_i} = \frac{1}{V_s} \int T^s n_i dS = -\frac{1}{3V_s} \sum_{l=1}^4 (T^s)^{(l)} n_{i(l)} S^{(l)}, \quad (4)$$

where V_s is the volume of the tetrahedral element, n_i is the outward normal unit vector of the infinitesimal surface dS , $n_{i(l)}$ is the outward normal unit vector of the triangular surface l of the tetrahedral element, $(T^s)^{(l)}$ is the rock matrix temperature of the node opposite the triangular surface l , and $S^{(l)}$ is the area of the triangular surface l .

By substituting Equation 4 into Equation 1, the heat flow rate in the i direction can be expressed as

$$q_i^s = k_{ij}^s \left(\frac{1}{3V_s} \sum_{l=1}^4 (T^s)^{(l)} n_{j^{(l)}} S^{(l)} \right), \quad (5)$$

where q_i^s is the rock matrix heat flow rate in the i direction and k_{ij}^s is the thermal conductivity of the rock matrix.

Therefore, the heat into node 1 from the tetrahedral element Q_{T1234}^s per unit time can be calculated as follows³⁰:

$$Q_{T1234}^s = -\frac{q_i^s n_{i^{(1)}} S^{(1)}}{3}, \quad (6)$$

where $n^{(1)}$ is the outer normal unit vector of the triangular surface $S^{(1)}$ opposite node 1.

The heat into node 1 from the other tetrahedral elements (T1354, T1564, T1624, T1237, T1357, T1567, and T1627) can be obtained similarly. The total heat into node 1 per unit time from all the tetrahedral elements that are connected to node 1 can be expressed as follows:

$$Q_{total}^s = Q_{T1234}^s + Q_{T1354}^s + Q_{T1564}^s + Q_{T1624}^s + Q_{T1237}^s + Q_{T1357}^s + Q_{T1567}^s + Q_{T1627}^s. \quad (7)$$

Equation 2 is then solved using the finite difference method, and the rock matrix temperature of node 1 at the next time step is given as follows:

$$T_{t+\Delta t}^s = T_t^s + \frac{Q_{total}^s}{C_s M} \Delta t. \quad (8)$$

The temperatures of the other discrete nodes can be obtained similarly. Thus, the temperature evolution of the continuum can be obtained.

Because an explicit algorithm is used in the heat conduction calculation of the continuum, the time step should be smaller than the critical time step to ensure the stability of the numerical calculation. The critical time step is given by

$$\Delta t_c = \frac{(\Delta x)^2}{4\kappa \left[1 + \frac{h\Delta x}{2k} \right]}, \quad (9)$$

where Δx is the size of the smallest element, h is the convection heat transfer coefficient, and κ is the thermal diffusivity ($\kappa = k/(\rho_s C_s)$ when $k_x = k_y = k_z$).

2.2 | Heat conduction in the fracture

When fluid flows in a fracture, heat is transported by conduction and convection. The fracture in the 3D FDEM is represented by the triangular surface of the joint element. Thus, fluid flow and fluid heat conduction and convection in the fracture are performed on the triangular surface of the joint element and are treated as a two-dimensional problem in FDEM-TH3D.

As shown in Figure 2A and 2B, a fracture surface in FDEM3D consists of triangular surfaces. One triangular surface constitutes a two-dimensional planar heat flow path. The temperature distribution on the fracture surface is represented by the temperatures of the nodes in Figure 2B. The calculation of the fluid temperature distribution on the fracture surface as shown in Figure 2B is introduced here.

Triangular surfaces $\Delta 123$, $\Delta 134$, $\Delta 145$, $\Delta 156$, $\Delta 167$, and $\Delta 178$ are connected to node 1. Taking the triangular surface $\Delta 123$ as an example, the temperatures of nodes 1, 2, and 3 are denoted T_1 , T_2 , and T_3 , respectively. In the local coordinate system (both the x and y axes are on the triangular surface $\Delta 123$, whereas the z -axis is perpendicular to the triangular surface), the coordinates of these three nodes are denoted $(x_n, y_n, 0)$ (n is the node number; $n = 1, 2, 3$). Assuming the temperature on the triangular surface has a linear distribution, the temperature gradient on the triangular surface is constant and can be expressed by

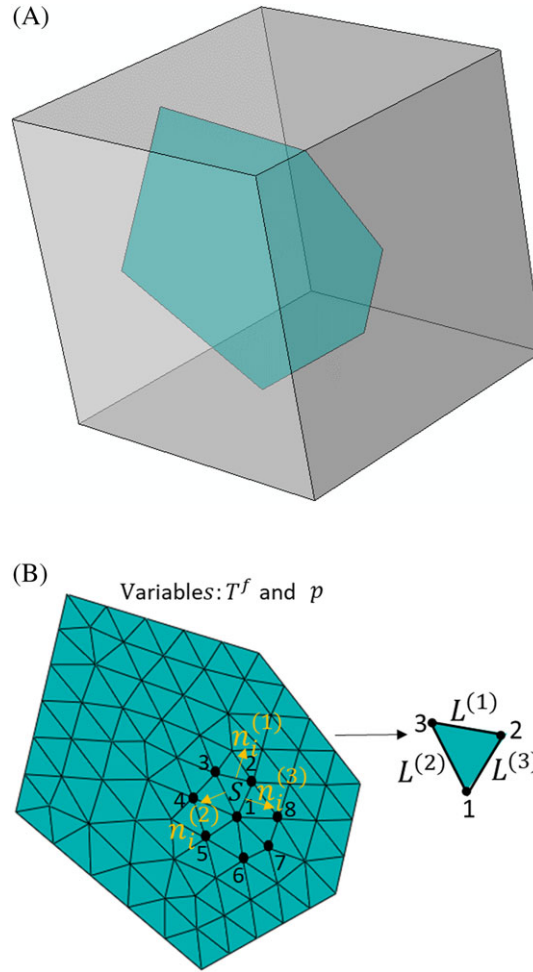


FIGURE 2 The mesh of the fracture surface in the FDEM-TH3D model for heat conduction and convection for fluid in fracture [Colour figure can be viewed at wileyonlinelibrary.com]

$$\frac{\partial T^f}{\partial x_i} = \frac{1}{S_s} \int \frac{\partial T^f}{\partial x_i} dS. \quad (10)$$

where T^f is the fluid pressure on the triangular surface and S is the area of the triangular surface $\Delta 123$.

According to the divergence theorem, Equation 10 can be written as

$$\frac{\partial T^f}{\partial x_i} = \frac{1}{S_s} \int T^f n_i dS = \frac{1}{S} \sum_{m=1}^3 \left(\overline{T^f} \right)^{(m)} \epsilon_{ij} \Delta x_j^{(m)}, \quad (11)$$

where A is the area of the triangular surface, n_i is the outward normal vector of the infinitesimal surface dS , $\left(\overline{T^f} \right)^{(m)}$ is the average fluid temperature of the edge m , $\Delta x_j^{(m)}$ is the difference in x_j between the two nodes of edge m , and ϵ_{ij} is a two-dimensional permutation tensor $\epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

According to Equation 1, the heat flow rate in the i direction can be expressed as follows:

$$q_i^f = -k_{ij}^f \frac{\partial T^f}{\partial x_j}, \quad (12)$$

where q_i^f is the fluid heat flow rate in the i direction, k_{ij}^f is the thermal conductivity coefficient tensor of the fluid, and T^f is the fluid temperature.

By substituting Equation 11 into Equation 12, the heat flow rates in the x and y directions (q_x^f and q_y^f) can be obtained.

The heat into node 1 from the triangular surface $\Delta 123$ per unit time can be calculated by

$$Q_{\Delta 123}^{f, \text{cond}} = -\frac{q_i^f n_{i(1)} L^{(1)}}{2}, \quad (13)$$

where $n_{i(1)}$ is the outer normal unit vector of the edge opposite node 1 and $L^{(1)}$ is the length of the edge opposite node 1.

Similarly, we can obtain the heat into node 1 from the other triangular surfaces ($\Delta 134$, $\Delta 145$, $\Delta 156$, $\Delta 167$, $\Delta 178$, and $\Delta 182$) that are directly connected to the node. Finally, the total heat into node 1 per unit time due to heat conduction in the fluid can be expressed by

$$Q^{f, \text{cond}} = Q_{\Delta 123}^{f, \text{cond}} + Q_{\Delta 134}^{f, \text{cond}} + Q_{\Delta 145}^{f, \text{cond}} + Q_{\Delta 156}^{f, \text{cond}} + Q_{\Delta 167}^{f, \text{cond}} + Q_{\Delta 178}^{f, \text{cond}} + Q_{\Delta 182}^{f, \text{cond}}. \quad (14)$$

If the heat convection in the fracture is not considered, according to Equation 2 and the finite difference method, the fluid temperature at node 1 at the next time step is given as follows

$$T_{t+\Delta t}^f = T_t^f + \frac{Q^{f, \text{cond}} \Delta t}{C_f \rho_f V_f}. \quad (15)$$

2.3 | Heat convection in the fracture

In addition to heat conduction, heat flow on a triangular surface may also be caused by heat convection. Again taking the triangular surface $\Delta 123$ as an example (Figure 2B), the temperature gradient on the triangular surface can also be expressed as

$$\nabla T = \left(\frac{\partial T^f}{\partial x}, \frac{\partial T^f}{\partial y} \right), \quad (16)$$

where $\frac{\partial T^f}{\partial x}$ and $\frac{\partial T^f}{\partial y}$ can be obtained from Equation 11.

The heat into node 1 from the triangular surface $\Delta 123$ due to heat convection is given by

$$Q_{\Delta 123}^{f, \text{conv}} = -\frac{1}{3} \rho_f C_f \mathbf{q}_f \cdot \nabla T A, \quad (17)$$

where $\mathbf{q}_f$ is the fluid flow rate vector on the triangular surface $\Delta 123$, which can be obtained as follows.

Assuming the fluid pressure p on the triangular surface has a linear distribution, the fluid pressure gradient on the surface is constant and can be expressed by

$$\frac{\partial p}{\partial x_i} = \frac{1}{A} \int_A \frac{\partial p}{\partial x_i} dA. \quad (18)$$

According to the divergence theorem, Equation 18 can be written as follows:

$$\frac{\partial p}{\partial x_i} = \frac{1}{A} \int_A p n_i ds = \frac{1}{A} \sum_{m=1}^3 \bar{p}^m \epsilon_{ij} \Delta x_j^m, \quad (19)$$

where A is the area of the triangular surface, n_i is the outward normal vector, $\bar{p}^m$ is the average fluid pressure at the edge m , Δx_j^m is the difference between the coordinate components of the two nodes of edge m , and ϵ_{ij} is the two-dimensional

permutation tensor $\epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

According to the cubic law, the fluid flow rate vector can be expressed by

$$\mathbf{q}_f = -\frac{a^3}{12\mu} \left(\frac{\partial p}{\partial x}, \frac{\partial p}{\partial y} \right), \quad (20)$$

where a is the hydraulic aperture of the fracture, μ is the fluid's dynamic viscosity, and p is the fluid pressure.

The fluid flow rates in the x and y directions are obtained by substituting Equation 19 into Equation 20; that is, the flow rate vector $\mathbf{q}_f$ is obtained.

The heat into node 1 of the fluid from the other triangular surfaces due to heat convection can be similarly obtained. Thus, the total heat induced by heat convection per unit time is given by $Q^{f_{conv}} = Q_{\Delta 123}^{f_{conv}} + Q_{\Delta 134}^{f_{conv}} + Q_{\Delta 145}^{f_{conv}} + Q_{\Delta 156}^{f_{conv}} + Q_{\Delta 167}^{f_{conv}} + Q_{\Delta 178}^{f_{conv}} + Q_{\Delta 182}^{f_{conv}}$.

According to the finite difference method and Equation 2, the temperature of node 1 at the next time step is given by

$$T_{t+\Delta t}^f = T_t^f + \frac{Q_{total}^f \Delta t}{C_f \rho_f V_f}, \quad (21)$$

where T_t^f and $T_{t+\Delta t}^f$ are the fluid temperatures of node 1 at the current and next time step, respectively, C_f is the specific heat of the fluid, ρ_f is the fluid density, $Q_{total}^f = Q^{f_{cond}} + Q^{f_{conv}}$ is the total heat into node 1 per unit time, Δt is the time step, and V_f is one-third of the volume of all the joint elements that are connected to node 1.

2.4 | Heat exchange between the fluid and rock

Because the temperatures of the rock matrix and the fluid are usually different, heat exchange between the rock matrix and the fluid occurs along the fracture. As shown in Figure 3, the temperatures of the rock matrix at the two walls of the fracture are T_+^s and T_-^s , respectively. The temperature of the fluid in the fracture is T^f . The area of the fracture is A , and the heat transfer coefficient between the rock matrix and the fluid is h . The heat exchange between the rock matrix and the fluid in the fracture per unit time is given by

$$Q^e = h(T_+^s - T^f)A + h(T_-^s - T^f)A. \quad (22)$$

Thus, the fluid temperature in the fracture is updated by

$$T_{t+\Delta t}^f = T_t^f + \frac{(Q^f + Q^e) \Delta t}{C_f \rho_f V_f}, \quad (23)$$

and the temperature of the rock matrix is finally updated by

$$T_{t+\Delta t}^s = T_t^s + \frac{(Q^s - Q^e) \Delta t}{C_s M}. \quad (24)$$

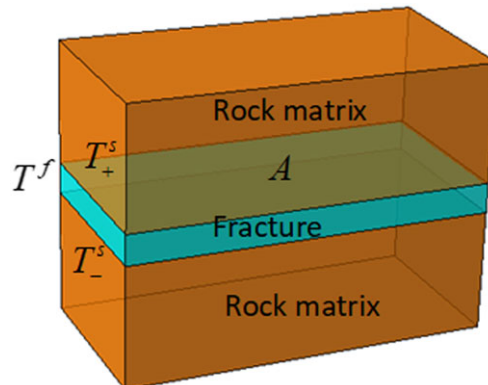


FIGURE 3 Heat exchange between the fluid in a fracture and the rock matrix [Colour figure can be viewed at wileyonlinelibrary.com]

2.5 | Effect of heat transport on fluid flow in the fractured rock

Fluid flow affects heat transport in the fractured rock through conduction and convection, and the heat transport affects the fluid temperature. A change in fluid temperature causes the fluid viscosity to change, which affects the flow in the fracture. The effect of fluid flow on heat transport is presented in Sections 2.2 to 2.4. The effect of heat transport on the fluid flow when the fluid is water can be reflected by the empirical formula.³³

$$\mu = \rho_f \left(\frac{0.01775}{10000 \left(1 + 0.033T^f + 0.000221(T^f)^2 \right)} \right), \quad (25)$$

where μ is the dynamic viscosity of the fluid (Pa/second) and T^f is the fluid temperature ($^{\circ}\text{C}$). This formula is not the same as that discussed in the literature.³⁴ However, the temperature dependences of the dynamic viscosity of water represented by these two formulas are very similar, as shown in Figure 4.

Substituting Equation 25 into Equation 20 shows that heat transport alters the fluid flow rate because heat transport may change the fluid temperature. The fluid flow rate increases when the fluid temperature increases. Thus, the effect of heat transport on the fluid flow is reflected in the 3D coupled hydrothermal model.

2.6 | Calculation procedure

The calculation procedure is shown in Figure 5. It contains the following three parts: the heat conduction calculation of the rock matrix, fracture seepage,³⁰ the fluid heat conduction and convection calculation, the heat exchange between the fluid and the rock matrix at the fracture, and updating the temperatures of the rock matrix and the fluid according to Equations 23 and 24.

3 | VERIFICATION OF THE 3D COUPLED HYDROTHERMAL MODEL

3.1 | Heat conduction of a fluid

As shown in Figure 6, a block with length $L = 2$ m contains a single fracture. The fracture is filled with fluid, but the fluid velocity is zero. The fluid temperature at the left boundary T_1^f is fixed at 100°C , whereas the fluid temperature at the right boundary T_2^f is fixed at 0°C . The other boundaries are adiabatic. The thermal conductivity of the rock matrix is ignored because this example only considers heat conduction in the fluid. The temperature distribution of the fluid in the fracture is determined by the 3D coupled hydrothermal model and compared with the analytical solution. The physical parameters include the fluid density $\rho_f = 1000$ kg/m³, the thermal conductivity coefficient of the fluid $k^f = 1.6$ W/m $^{\circ}\text{C}$, and the specific heat of the fluid $C_f = 0.2$ J/kg $^{\circ}\text{C}$.

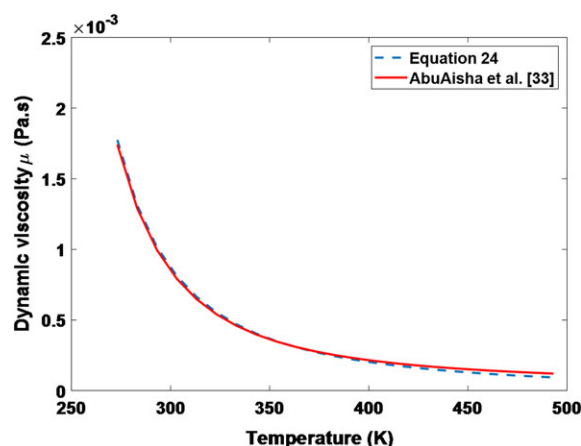


FIGURE 4 Effect of temperature on the dynamic viscosity of water [Colour figure can be viewed at wileyonlinelibrary.com]

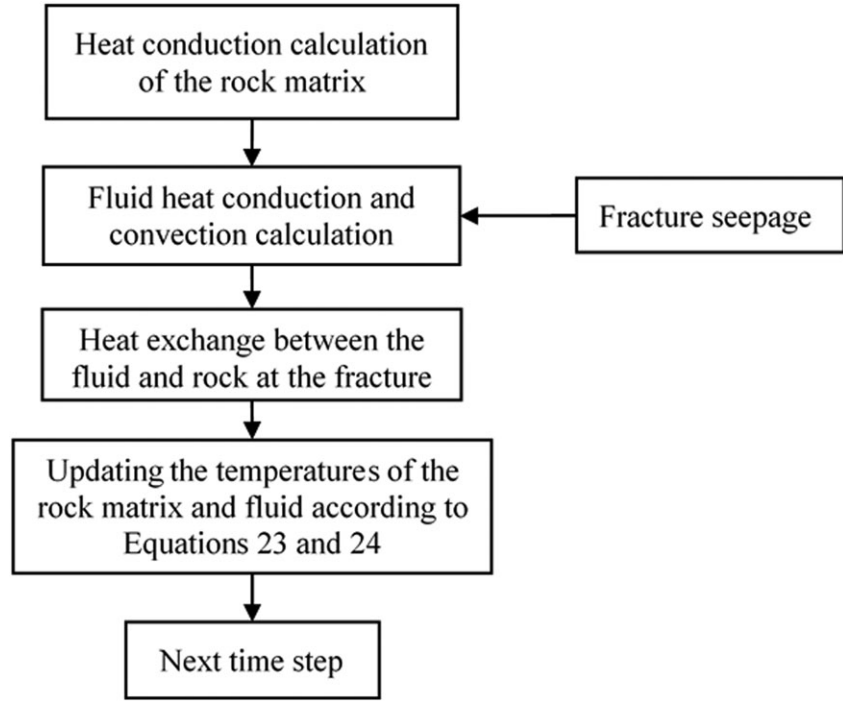


FIGURE 5 Calculation procedure of the 3D coupled hydrothermal model FDEM-TH3D

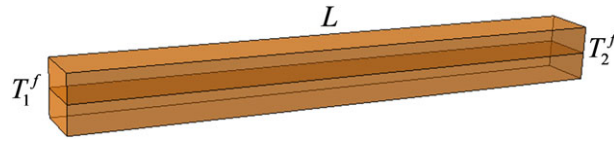


FIGURE 6 Heat conduction of fluid in a single fracture [Colour figure can be viewed at wileyonlinelibrary.com]

The analytical solution to the problem is given by^{35,36}

$$T^f(x, t) = T_1^f + \frac{x}{L} (T_2^f - T_1^f) + \frac{2}{\pi} \sum_{n=1}^{\infty} e^{-kn^2\pi t/L^2} \left(\frac{T_2^f \cos(n\pi) - T_1^f}{n} \right) \sin \frac{n\pi x}{L}, \quad (26)$$

where T_1^f is the fluid temperature at the left boundary, T_2^f is the fluid temperature at the right boundary, t is the time, x is the distance to the left boundary, and the diffusivity $\kappa = k^f/(\rho_f C_f)$, where k^f is the thermal conductivity coefficient of the fluid, ρ_f is the fluid density, and C_f is the specific heat of the fluid.

Figure 7 presents the temperature distributions of the fluid in the fracture at different times. The numerical solutions agree well with the analytical solutions, which verify the accuracy of the coupled hydrothermal model in dealing with heat conduction in a fluid.

3.2 | Steady-state heat conduction and convection in a fluid

The example presented in Section 3.1 is also used in this example, and the following conditions are considered. In case 1, the fluid pressure at the left boundary is $P_1 = 2 \times 10^4$ Pa, whereas the fluid pressure at the right boundary is $P_2 = 0$ Pa. In case 2, the fluid pressure at the left boundary is $P_1 = 0$ Pa, but the fluid pressure at the right boundary is fixed at $P_2 = 2 \times 10^4$ Pa. The example shown in Section 3.1 is called case 0. Heat is transported by convection and conduction because the fluid flows in the fracture due to the pressure difference between the left and right boundaries. The coupled hydrothermal model is used to determine the temperature distribution of the fluid in the fracture when the fluid flow and heat transfer reach stability.

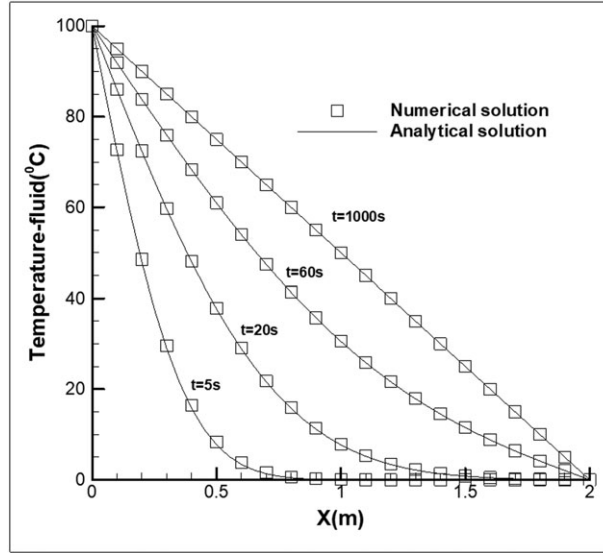


FIGURE 7 Temperature distributions of the fluid for the problem of heat conduction of fluid in a single fracture at $t = 5, 20, 60$, and 1000 seconds

The analytical solution to this problem is given by³⁷

$$\hat{T} = \hat{T}_2 \frac{e^{P_e \hat{x}} - 1}{e^{P_e} - 1}, \quad (27)$$

where $\hat{T} = (T^f - T_1^f)/T_1^f$, $\hat{T}_2 = (T_2^f - T_1^f)/T_1^f$, $\hat{x} = x/L$, and P_e is the Peclet number, which is defined by

$$P_e = \rho_f C_f q_f L / k^f, \quad (28)$$

where ρ_f is the fluid density, C_f is the specific heat of the fluid, q_f is the flow rate, and k^f is the thermal conductivity coefficient of the fluid.

The calculation parameters include the fluid density $\rho_f = 1000 \text{ kg/m}^3$, the thermal conductivity coefficient of the fluid $k^f = 1.6 \text{ W/m } ^\circ\text{C}$, the specific heat of the fluid $C_f = 0.2 \text{ J/kg } ^\circ\text{C}$, the dynamic viscosity of the fluid $= 0.001 \text{ Pa second}$, and the fracture aperture $a = 1 \times 10^{-4} \text{ m}$.

The temperature distributions obtained by the 3D coupled hydrothermal model are shown in Figures 8 and 9, and they agree well with the analytical solutions. In case (1), heat convection promotes heat transport because the fluid flow direction is the same as that of the heat conduction. In case (2), the heat convection inhibits heat transport because the fluid flow direction is opposite to that of the heat conduction.

3.3 | Transient heat conduction and convection in a fluid

The example in Section 3.1 is used again in this section. The fluid pressure at the left boundary is 10 Pa , and the pressure at the right boundary is 0 Pa . The initial temperature of the fluid in the fracture is 0°C . The fluid temperature at the left boundary T_1^f is fixed at 100°C . Both heat conduction and convection of the fluid are considered. The example is a one-dimensional convection and diffusion problem. The analytical solution to the problem is given by the following expression^{38,39}:

$$T^f(x, t) = \frac{T_1^f}{2} \left(\operatorname{erfc} \left(\frac{x - v_x t}{2\sqrt{Dt}} \right) + e^{\frac{xy_k}{D}} \operatorname{erfc} \left(\frac{x + v_x t}{2\sqrt{Dt}} \right) \right), \quad (29)$$

where T_1^f is the fluid temperature at the left boundary, $\operatorname{erfc}()$ is the complementary error function, x is the distance to the

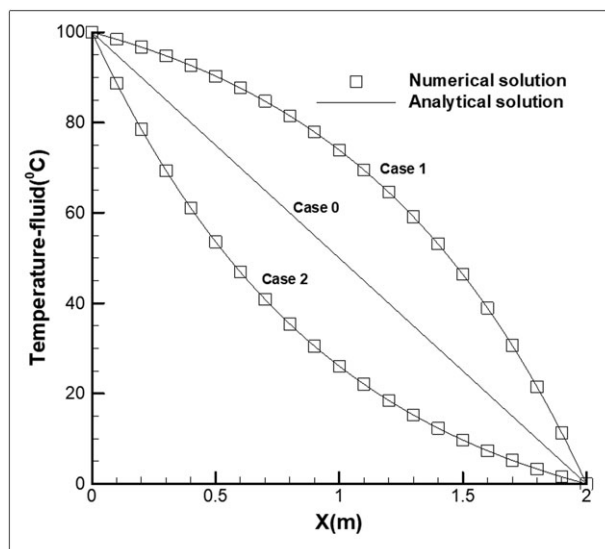


FIGURE 8 Temperature distributions of the fluid in the fracture for case 0 (pure heat conduction, $P_1 = P_2$), case 1 (heat conduction and convection, $P_1 > P_2$), and case 2 (heat conduction and convection, $P_1 < P_2$)

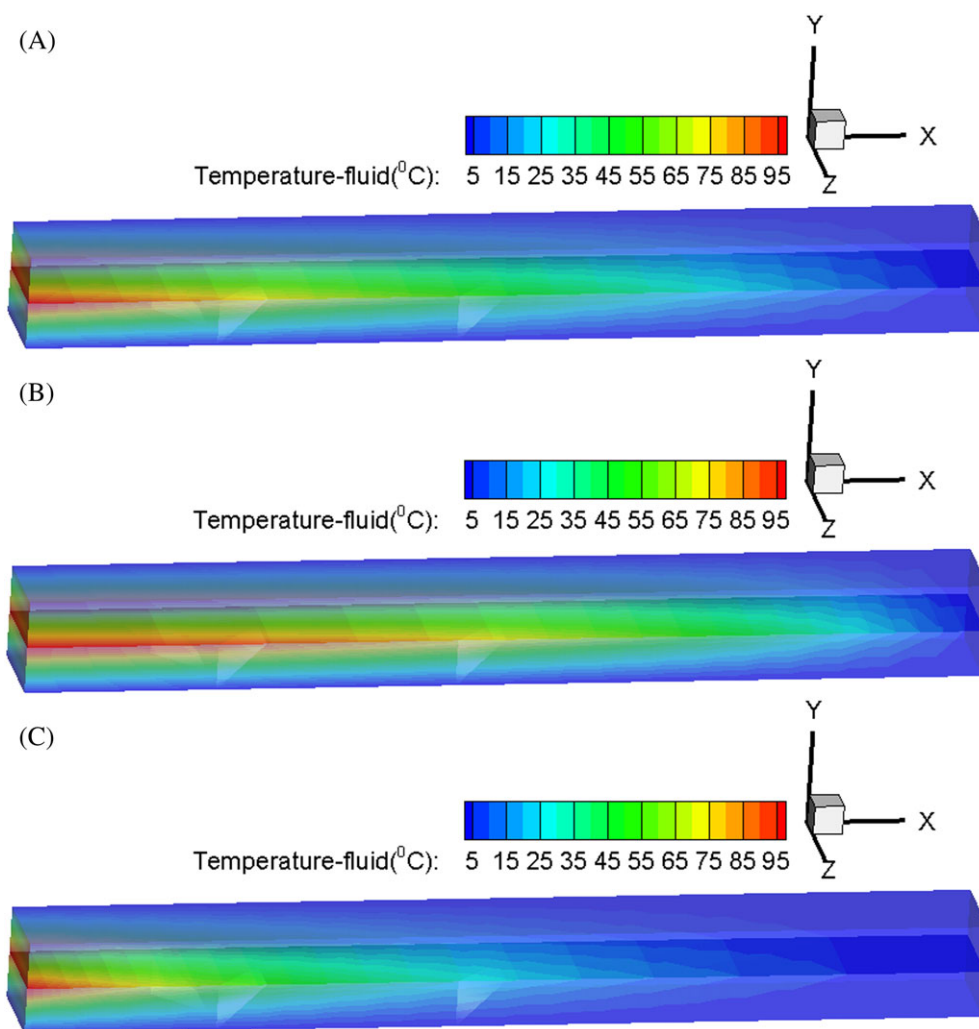


FIGURE 9 Temperature distributions of the fluid in the fracture for three cases corresponding to Figure 8 [Colour figure can be viewed at wileyonlinelibrary.com]

left boundary, v_x is the fluid velocity, t is the time, D is the thermal diffusivity ($D = \frac{k^f}{\rho_f C_f}$), k^f is the thermal conductivity coefficient of the fluid, ρ_f is the fluid density, and C_f is the specific heat of the fluid.

The 3D coupled hydrothermal model is used to determine the temperature evolution of the fluid in the fracture. The calculation parameters include the fluid density $\rho_f = 1000 \text{ kg/m}^3$, the thermal conductivity coefficient of the fluid $k^f = 0.6 \text{ W/m } ^\circ\text{C}$, the specific heat of the fluid $C_f = 4200 \text{ J/kg } ^\circ\text{C}$, the dynamic viscosity of the fluid $\mu = 0.001 \text{ Pa second}$, and the fracture aperture $a = 1 \times 10^{-4} \text{ m}$. The model is discretized into 5782 tetrahedral elements and 12 236 joint elements. The time step is $1 \times 10^{-1} \text{ second}$.

The temperature distribution of the fluid in the fracture calculated by the 3D coupled hydrothermal model is shown in Figure 10, and it agrees well with the analytical solution. The accuracy of the 3D coupled hydrothermal model in simulating heat conduction and convection of fluid in a fracture is therefore verified. Moreover, we studied the effect of the mesh size on the calculation results. As shown in Figure 11, the element size has an influence on the numerical results. However, when the mesh size is 0.05 m, the numerical result agrees well with the analytical solution.

3.4 | Heat exchange between the fluid and the solid

As shown in Figure 12A, we consider a block with dimensions of 4 m by 2 m. The block has an initial temperature T_i^s and is placed in a fluid with a constant temperature T^f .

The analytical solution to this problem is given by^{35,36}

$$T^s(x, y, t) = T^f + \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{e^{-\alpha_x r_{mn}^2 t}}{N(\beta_m) \cdot N(r_n)} \cos(\beta_m \cdot x) \cos(\gamma_n y) \cdot (T_i^s - T^f) \cdot \frac{\sin(\beta_m L_1) \cdot \sin(r_n L_2)}{\beta_m \cdot r_n}, \quad (30)$$

where

$$\alpha_x = \frac{k_x^s}{k_y^s} \cdot \frac{1}{\rho C_s}, \quad r_{mn}^2 = \beta_m^2 + \frac{k_y^s}{k_x^s} \cdot r_n^2, \quad \frac{1}{N(\beta_m)} = 2 \frac{\beta_m^2 + H_1^2}{L_1(\beta_m^2 + H_1^2) + H_1},$$

$$\frac{1}{N(r_n)} = 2 \frac{r_n^2 + H_2^2}{L_2(r_n^2 + H_2^2) + H_2}, \quad H_1 = \frac{h}{k_x^s}, \text{ and } H_2 = \frac{h}{k_y^s}.$$

The parameters β_m and r_n are the roots of the two transcendental equations

$$\begin{aligned} \beta_m \tan(\beta_m L_1) - H_1 &= 0 \\ r_n \tan(r_n L_2) - H_2 &= 0, \end{aligned} \quad (31)$$

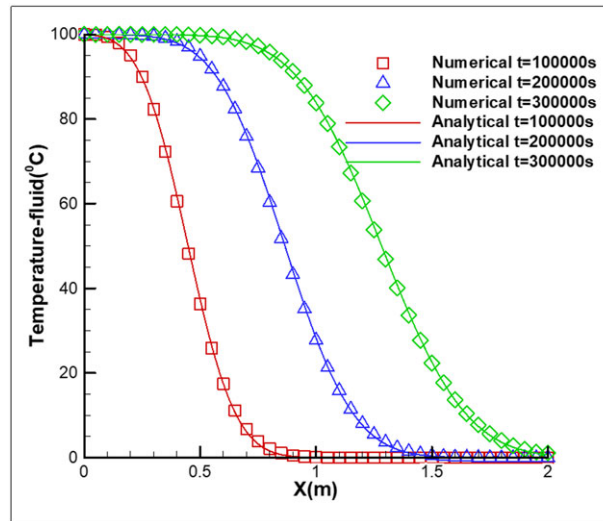


FIGURE 10 Temperature distributions of fluid in fractures during transient heat conduction and convection [Colour figure can be viewed at wileyonlinelibrary.com]

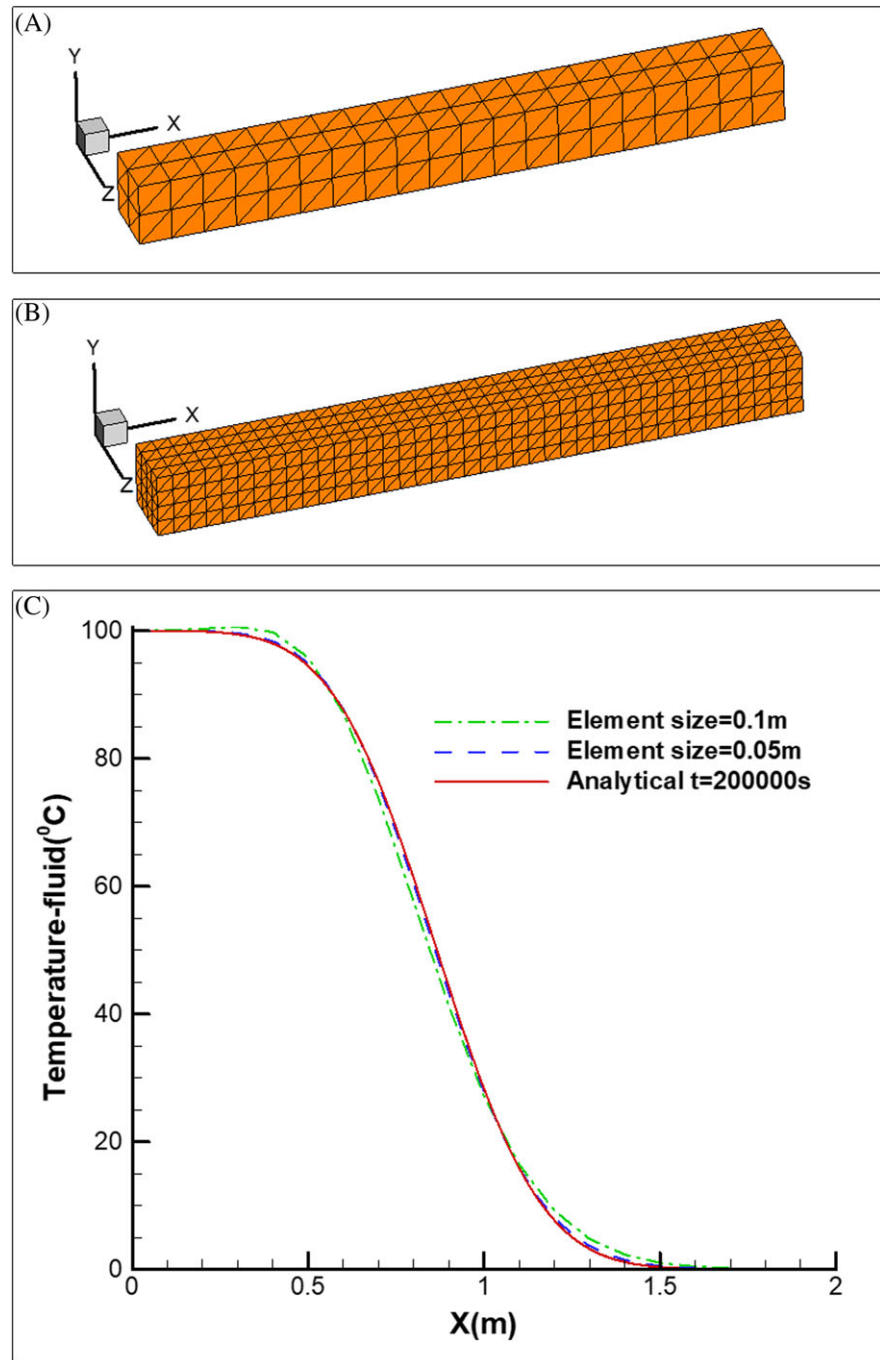


FIGURE 11 Temperature distributions of the fluid in the fracture for different element sizes [Colour figure can be viewed at wileyonlinelibrary.com]

respectively, where k_x^s and k_y^s are the thermal conductivities of the block in the x and y directions, respectively, ρ_s is the density of the block, C_s is the specific heat of the block, and t is time.

The temperature evolution of the block is then determined using the 3D coupled hydrothermal model. Due to the symmetry of the problem, we select the upper right corner of the model (exactly one-fourth of the entire model) for the analysis. Both the left and bottom sides of the one-fourth model are adiabatic boundaries, as shown in Figure 12 B. The calculation mesh is shown in Figure 12C. Two monitoring lines at $x = 1$ m and $y = 0.5$ m are set to determine the temperature evolution of the block, as shown in Figure 12B.

The calculation parameters used in the coupled hydrothermal model include the density of the block $\rho_s = 2760$ kg/m³, the specific heat of the block $C_s = 4200$ J/(kg °C), the thermal conductivity coefficients of the block in the x and y directions $k_x^s = 2$ W/(m °C) and $k_y^s = 1$ W/(m °C), respectively, and the heat transfer coefficient between the fluid and the block $h = 2$ W/(m² °C). The fluid temperature is fixed at $T^f = 30^\circ\text{C}$, and the initial temperature of the block $T_i^s = 300^\circ\text{C}$.

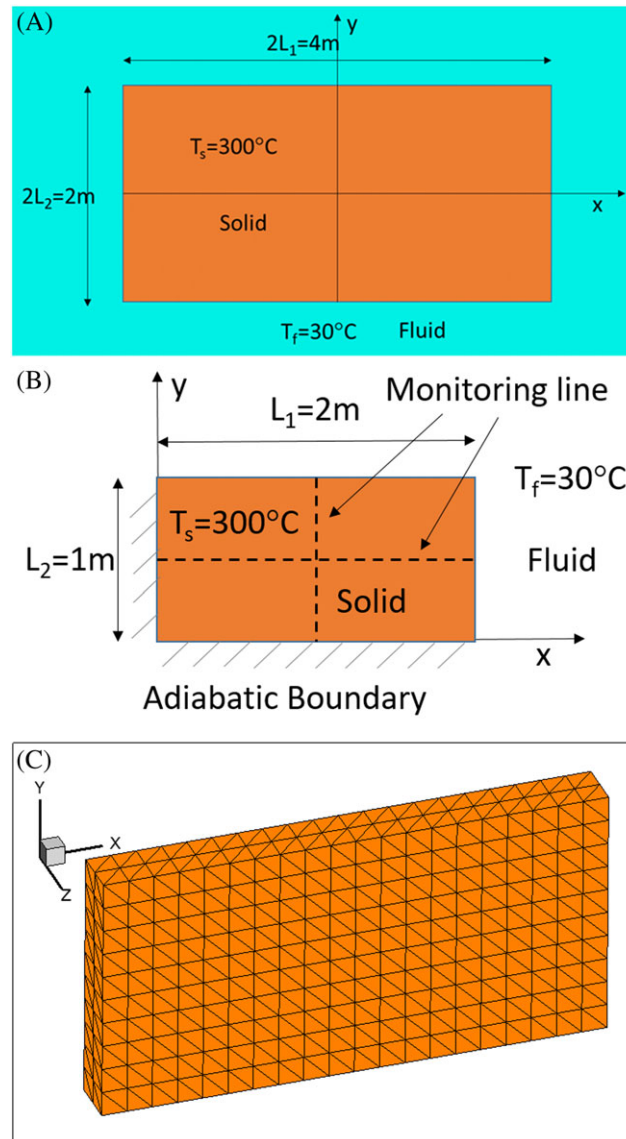


FIGURE 12 The computation model for heat exchange between the fluid and the solid [Colour figure can be viewed at wileyonlinelibrary.com]

The temperature evolutions on the monitoring lines obtained by the 3D coupled hydrothermal model are shown in Figures 13 and 14. The numerical solutions agree well with the analytical solution, proving the accuracy of the 3D coupled hydrothermal model.

4 | EXPERIMENTAL VERIFICATION EXAMPLE

4.1 | Model and parameters

The 3D coupled hydrothermal model is used to simulate the experiment of heat transport and fluid flow during heat exchange between a fluid and rock performed in a previous study.⁴⁰ In addition, the effects of the fracture aperture, fluid viscosity, and inlet-outlet pressure difference on the heat exchange between the fluid and the rock are investigated.

As shown in Figure 15, a cylindrical granite specimen with a diameter of 51 cm and a height of 102 cm is cut into two pieces along the radial direction of the cylinder to form a rectangular fracture surface. Low-temperature fluid is then injected from the bottom boundary and flows through the fracture surface. The temperatures on the outer

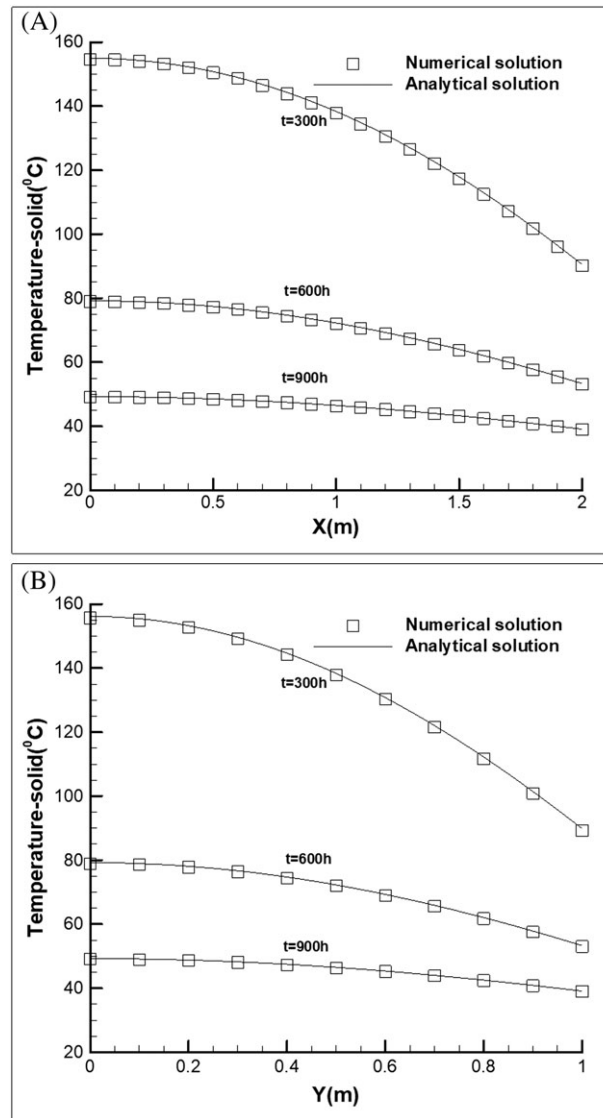


FIGURE 13 Temperature distributions in the block along two monitoring lines

boundaries of the cylindrical specimen are fixed at 100°C using a heating device. After the fluid flow and heat transport reach equilibrium, the fluid temperatures at the inlet and outlet are denoted as T_1^f and T_2^f , respectively. A monitoring line at the center of the fracture surface is set to determine the temperature evolution.

The 3D coupled hydrothermal model developed in this paper is used to simulate the experiment for 19 different conditions (rock specimen with a constant temperature of 100°C at the outer boundaries) as shown in Table 1; some of the experimental data and results in Zhao and Tso³⁹ are also presented in the left part of the table. In addition, two heat transfer coefficients are used. Thus, a total of 38 simulations are performed. The time step is 1×10^{-3} seconds. The total numbers of tetrahedral elements and joint elements are 1482 and 3163, respectively.

The calculation parameters include the fluid density $\rho_f = 1000 \text{ kg/m}^3$, the dynamic viscosity coefficient of the fluid $\mu = 0.001 \text{ Pa second}$, the specific heat of the fluid $C_f = 4200 \text{ J/(kg } ^\circ\text{C)}$, the thermal conductivity coefficient of the fluid $k_f = 0.6 \text{ W/(m } ^\circ\text{C)}$, the density of the rock specimen $\rho_s = 2360 \text{ J/(kg } ^\circ\text{C)}$, the specific heat of the rock specimen $C_s = 1015 \text{ J/(kg } ^\circ\text{C)}$, and the thermal conductivity coefficient of the rock specimen $k_s = 2 \text{ W/(m } ^\circ\text{C)}$. The other parameters (fracture aperture, fluid velocity, and fluid temperature at the inlet) are obtained directly from the experimental data,⁴¹ as shown in Table 1. Two values for the heat exchange coefficient h are used: (1) the average heat transfer coefficient and (2) the coefficient calculated according to the empirical formula $h = 543v^{0.9}$ in Zhao and Tso,³⁹ where v is the fluid flow velocity. Several simulation results are shown in the right part of Table 1 and Figures 16–21.

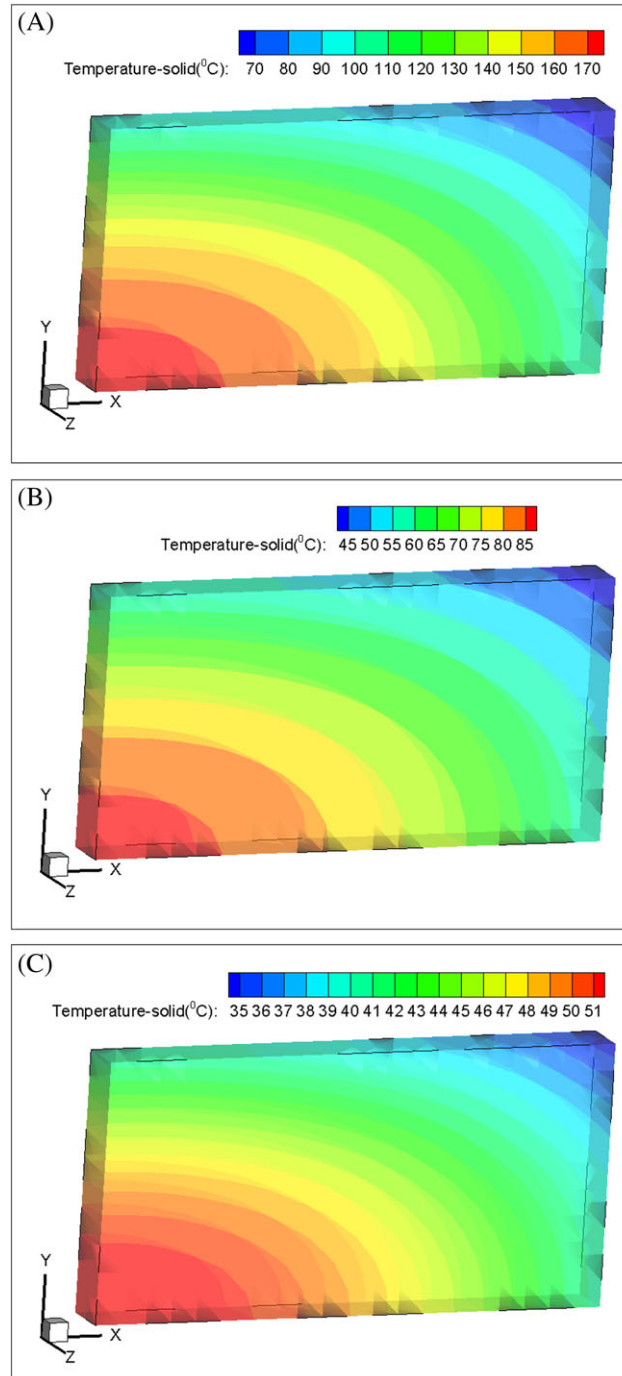


FIGURE 14 Temperature distributions in the block at $t = 300, 600$, and 900 hours [Colour figure can be viewed at wileyonlinelibrary.com]

4.2 | Numerical simulation results

4.2.1 | Numerical and experimental results

As shown in Figures 16 and 17, the numerical results agree well with the experimental results when the two different heat exchange coefficients are used. Both errors of the fluid temperature at the outlet are no more than 9% compared with the experimental results. Thus, the accuracy of the 3D coupled hydrothermal model has been further verified. An important reason for the error is that the fracture surface in the experiment is rough and wavy,⁴¹ which is different from the fracture surface in the numerical model with a constant aperture. Another possible reason is that the accuracy of the experimental data may be limited.²

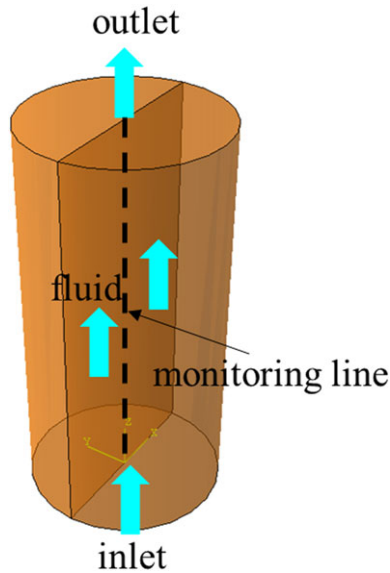


FIGURE 15 Cylindrical granite specimen used for the experiments of heat exchange between a fluid and rock [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 1 Experimental data and results⁴⁰ and numerical results from the 3D coupled hydrothermal model presented in this paper

Experimental Data and Results ⁴⁰					Numerical Results			
Average Fracture Aperture a , mm	Flow Velocity, v , mm/s	Inlet Temperature, T_1 , °C	Average Heat Transfer Coefficient, $h1$, W/(m ² °C)	Outlet Temperature, T_2 , °C	Outlet Temperature, T_2 , °C, $h = h1$	Error ($h = h1$)	Outlet Temperature, T_2 , °C, $h = 543v^{0.9}$	Error ($h = 543v^{0.9}$)
15.85	42.25	65	24.04	96	94.0064	6.43%	96.4069	1.31%
15.85	60.86	64	37.7	96	94.655	4.20%	95.8543	0.46%
15.85	82.3	62	56.04	96	95.207	2.33%	95.4115	1.73%
15.85	102.57	60	76.99	96	93.4909	6.97%	94.5354	4.07%
15.85	128.2	59	100.7	95	95.9293	2.58%	94.5354	1.29%
16.44	40.3	62	23.14	95	93.1871	5.49%	95.8324	2.52%
16.44	60.02	59	39.01	95	94.0329	2.69%	95.0591	0.16%
16.44	66.09	59	43.95	95	94.2608	2.05%	94.9489	0.14%
16.44	78.6	58	53.36	95	94.3075	1.87%	94.6199	1.03%
16.98	25.88	61	16.57	97	93.8911	8.64%	95.8817	3.11%
16.98	53.73	57	36.84	96	93.987	5.16%	94.6167	3.55%
16.98	59.46	57	41.7	96	94.2167	4.57%	94.4943	3.86%
16.98	70.24	56	51.77	96	94.5749	3.56%	94.1576	4.61%
16.98	78.62	55	60.39	96	94.8429	2.82%	93.8798	5.17%
21.16	22.4	54	16	94	91.2893	6.78%	92.7235	3.19%
21.16	32.78	54	24.45	94	91.9575	5.11%	92.2968	4.26%
21.16	42.03	53	32.84	94	92.3032	4.14%	91.6739	5.67%
21.16	63.5	50	56.02	94	93.3283	1.53%	90.5192	7.91%
21.16	77.45	46	71.49	93	93.3089	0.66%	89.4433	7.57%

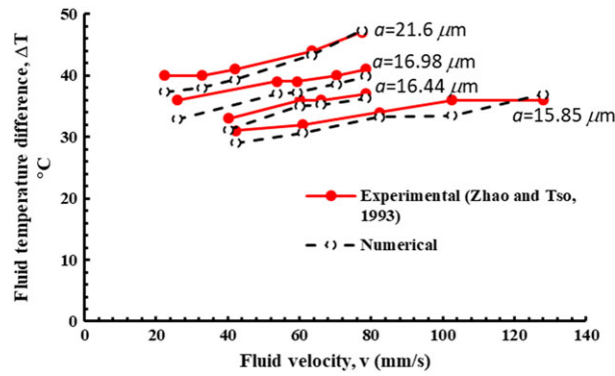


FIGURE 16 Effect of fluid velocity on the fluid temperature difference for model in Figure 15 when the heat transfer coefficient h = average heat transfer coefficient [Colour figure can be viewed at wileyonlinelibrary.com]

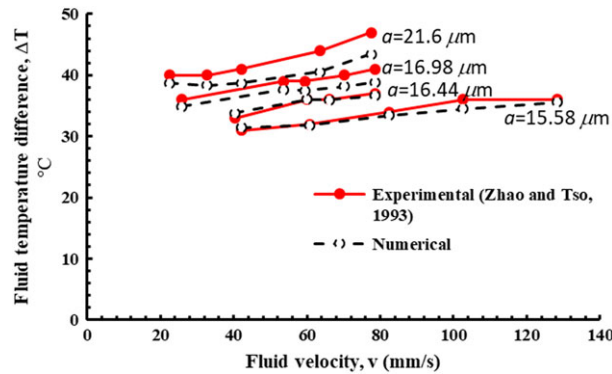


FIGURE 17 Effect of the fluid velocity on the fluid temperature difference for model in Figure 15 when the heat transfer coefficient $h = 543v^{0.9}$ [Colour figure can be viewed at wileyonlinelibrary.com]

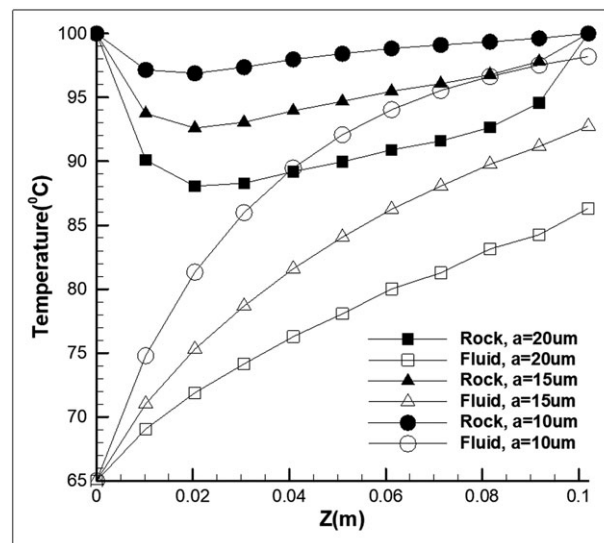


FIGURE 18 Temperature distributions in the fluids and rock along the monitoring line at $t = 1.4$ hours for different values of the fracture aperture a

4.2.2 | Effect of fracture aperture on heat exchange between the fluid and the rock

The temperature distributions along the monitoring line at $t = 1.4$ hours when the fracture apertures are $a = 10, 15$, and $20 \mu\text{m}$ are shown in Figure 18. The temperature of the rock matrix at the monitoring line decreases with the fracture aperture, which means that the increasing fracture aperture promotes heat transfer between the fluid and the rock.

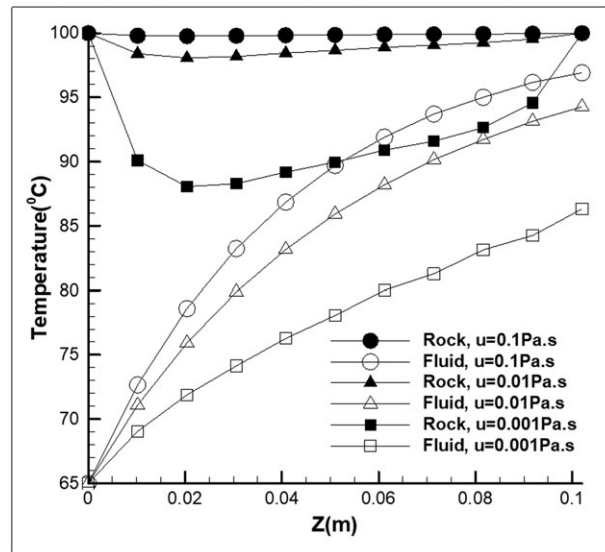


FIGURE 19 Temperature distributions in the fluids and rock along the monitoring line for different values of the fluid viscosity μ

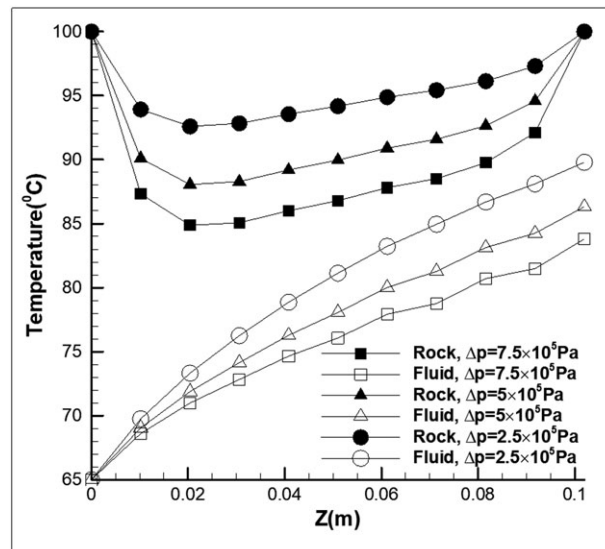


FIGURE 20 Temperature distributions in the fluid and rock along the monitoring line for different values of the fluid pressure difference Δp

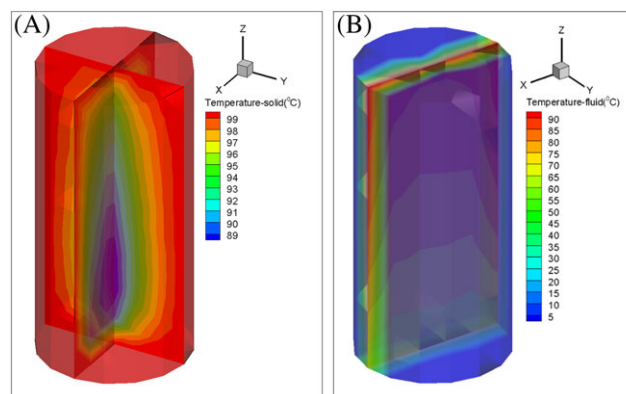


FIGURE 21 Temperature distributions in the rock and fluid along the fracture surface ($a = 20 \mu\text{m}$, $\mu = 0.001 \text{ Pa second}$, $\Delta p = 5 \times 10^5 \text{ Pa}$)
[Colour figure can be viewed at wileyonlinelibrary.com]

However, the fluid temperature at the monitoring line decreases with increasing fracture aperture. The fluid velocity increases with increasing fracture aperture; therefore, the fluid in the fracture with large aperture is not heated sufficiently and flows out from the bottom boundary, which results in the fluid in the fracture having a lower temperature.

4.2.3 | Effect of fluid viscosity on heat exchange between the fluid and the rock

To study the effect of the fluid viscosity on the heat transfer between the fluid and the rock, fluid dynamic viscosities of $\mu = 0.001, 0.01$, and 0.1 Pa second are used. As shown in Figure 19, the temperature of the rock matrix at the monitoring line increases with the dynamic viscosity of the fluid, indicating that increasing the fluid viscosity inhibits the heat exchange between the fluid and the rock. However, the fluid temperature at the monitoring line increases with increasing fluid viscosity. The fluid velocity decreases with increasing fluid viscosity; therefore, the high-viscosity fluid is heated sufficiently before it flows out from the top boundary.

4.2.4 | Effect of the fluid pressure difference on the heat transfer between the fluid and the rock

As shown in Figure 20, the temperature of the rock matrix at the monitoring line decreases with an increase in the fluid pressure difference between the inlet and outlet, which means that increasing the fluid pressure difference promotes heat exchange between the fluid and the rock. However, the fluid temperature at the monitoring line decreases with increasing fluid pressure difference; this occurs because the fluid velocity increases with increasing fluid pressure difference, so the fluid in the fracture with a high pressure difference is not heated sufficiently and flows out from the top boundary.

Figure 21 shows the temperature distributions in the rock specimen and fluid in the fracture. The fluid temperature increases gradually from the inlet to the outlet, whereas the temperature of the granite specimen is lowest near the inlet.

5 | APPLICATION: HEAT TRANSPORT AND FLUID FLOW IN ROCK WITH A FRACTURE NETWORK

5.1 | Calculation model and parameters

As a simple application of the 3D coupled hydrothermal model, we simulate heat transport and fluid flow in the cuboid containing a three-dimensional fracture network shown in Figure 22. The cuboid has dimensions of $2 \text{ m} \times 1 \text{ m} \times 0.2 \text{ m}$ and contains nine fractures (persistent fractures J1-J8 and nonpersistent fracture J9). It should be noted that fracture J7 is not perpendicular to the front and rear surfaces of the cuboid. Therefore, this problem cannot be simplified as a two-dimensional problem but rather is a true three-dimensional problem. Fluid is injected at the left boundary (inlet) and flows out from the right boundary (outlet). This example only considers the coupled hydrothermal process.

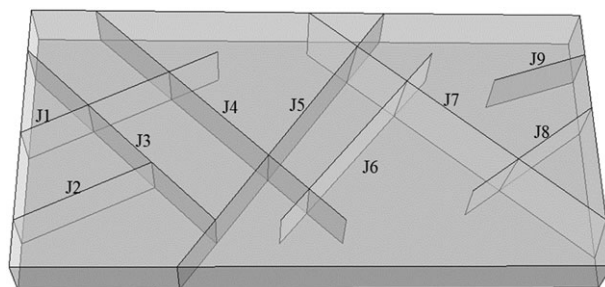


FIGURE 22 Cuboid containing a three-dimensional fracture network

The following boundary conditions are used: The fluid pressure at the left boundary is fixed at 200 kPa, whereas that at the right boundary is fixed at 100 kPa, and the other boundaries are impermeable. The fluid temperature at the left boundary is fixed at 20°C, whereas all the boundaries of the cuboid are thermally insulated. The initial temperature of the cuboid and fluid is 100°C.

The calculation parameters include the fluid density $\rho_f = 1000 \text{ kg/m}^3$, the viscosity coefficient $\mu = 1 \text{ mPa second}$, the specific heat of the fluid $C_f = 4200 \text{ J/(kg } ^\circ\text{C)}$, the thermal conductivity of the fluid $k^f = 0.6 \text{ W/(m } ^\circ\text{C)}$, the fracture aperture $a = 1 \times 10^{-4} \text{ m}$, the density of the rock matrix $\rho_s = 2360 \text{ kg/m}^3$, the specific heat of the rock matrix $C_s = 1015 \text{ J/(kg } ^\circ\text{C)}$, the thermal conductivity of the rock matrix $k^s = 0.5 \text{ W/(m } ^\circ\text{C)}$, and the heat exchange coefficient between the fluid and the rock $h = 900 \text{ W/(m}^2 \text{ } ^\circ\text{C)}$. The time step is 0.1 second.

5.2 | Numerical simulation results

5.2.1 | Seepage field

As shown in Figure 23A, because the nonpersistent fracture (J9) does not connect with the left boundary, the fluid pressure in it is uniform, and the fluid does not flow. As shown in Figure 23B, the net flow of fluid in the nonpersistent fracture is zero. Thus, the fluid in the nonpersistent fracture cannot perform heat transfer through fluid convection. The nonpersistent fracture does not have an obvious effect on the heat transport in the fractured rock. The temperature distribution in Figure 24 confirms this explanation.

5.2.2 | Temperature field of the rock matrix

As the cold fluid is injected into the high-temperature fractured rock, heat is transferred from the rock to the fluid in the fractures and carried away by the fluid through heat convection. The temperature of the rock matrix near the persistent fractures, where the fluid velocity is large, decreases rapidly, whereas the temperature of the rock matrix far from the fractures decreases slowly. As shown in Figure 24, at 2.8 to 61.1 hours, the cold front moves preferentially along the persistent fractures J1, J1, and J3 and reaches the right boundary of the cuboid via the persistent fractures J4, J5, J6, J7, and J8. The persistent fractures form fluid flow channels. The nonpersistent fracture J9 does not affect the heat transfer in the cuboid, as shown in Figure 24 at 38.9 hours, because static fluids cannot pass heat through convection (the net flow

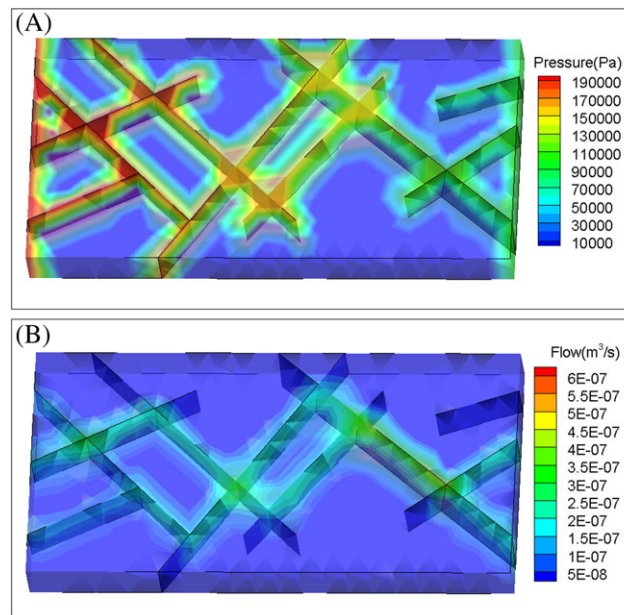


FIGURE 23 Fluid pressure distribution and net fluid flow distribution during stable seepage [Colour figure can be viewed at wileyonlinelibrary.com]

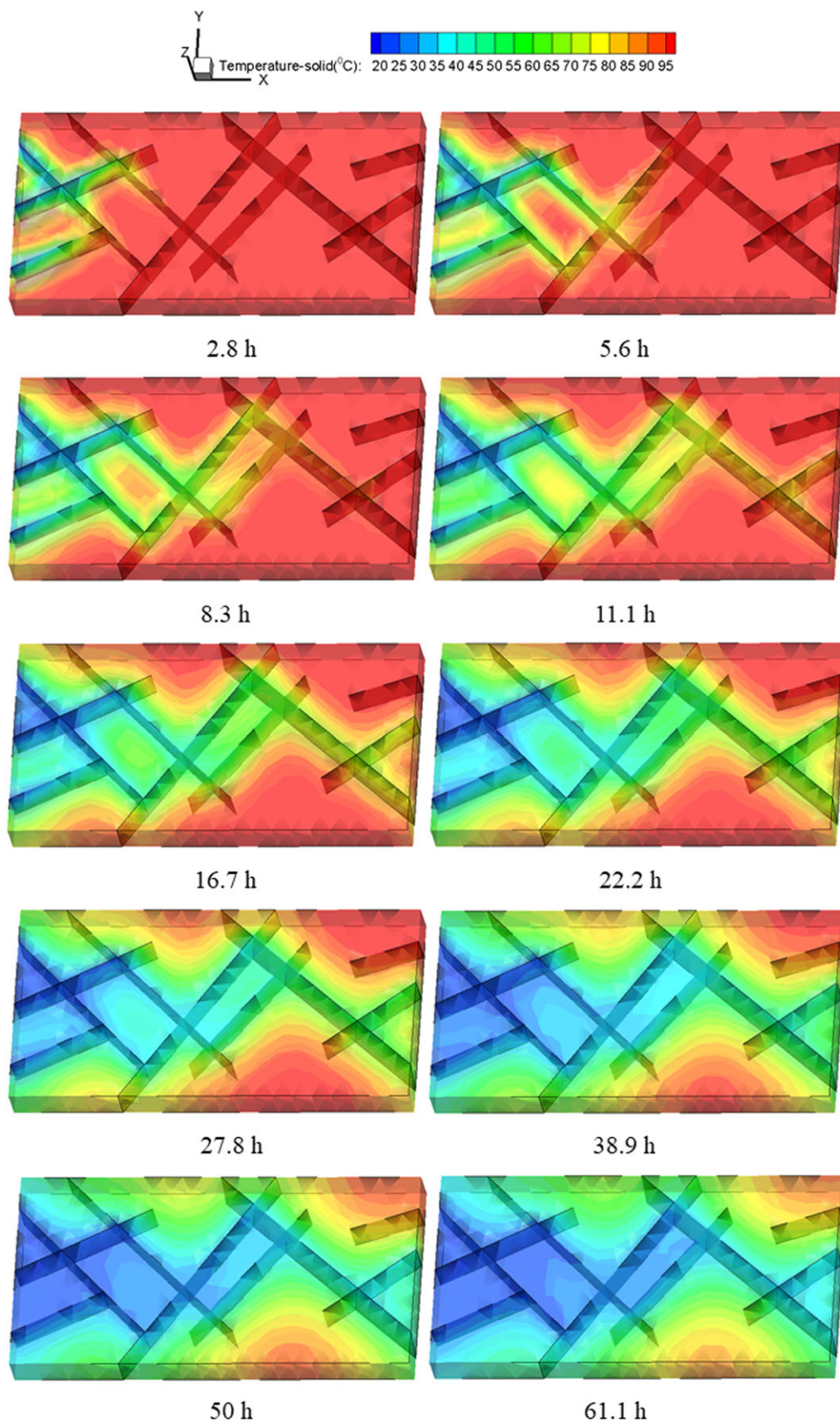


FIGURE 24 Temperature distributions in the rock matrix at different times [Colour figure can be viewed at wileyonlinelibrary.com]

of the fluid in the nonpersistent fracture is zero; ie, the fluid velocity is zero (see Figure 23B)). The cold front moves very quickly and reaches the right boundary of the cuboid, which constitutes a premature thermal breakthrough⁴² (ie, the fluid temperature at the outlet is very low, but many areas are still at high temperature). This is detrimental to the exploitation of geothermal energy, as shown in Figure 24 at 27.8 hours. With the continued injection of fluid, although the fluid temperature at the right boundary is already low, there are still several zones in the cuboid that remain at high temperature, as shown in Figure 24 at 50 hours. The continued heat supply from those zones prevents the fluid temperature at the outlet from decreasing rapidly but produces a gradual decrease over a long period of time, which is the so-called long tail effect.⁴² If an equivalent continuum approach was to be used to model the heat transport and fluid flow in fractured rock without considering the fracture network, the simulation results would not reflect the effect of the fractures on the fluid flow and heat transport.

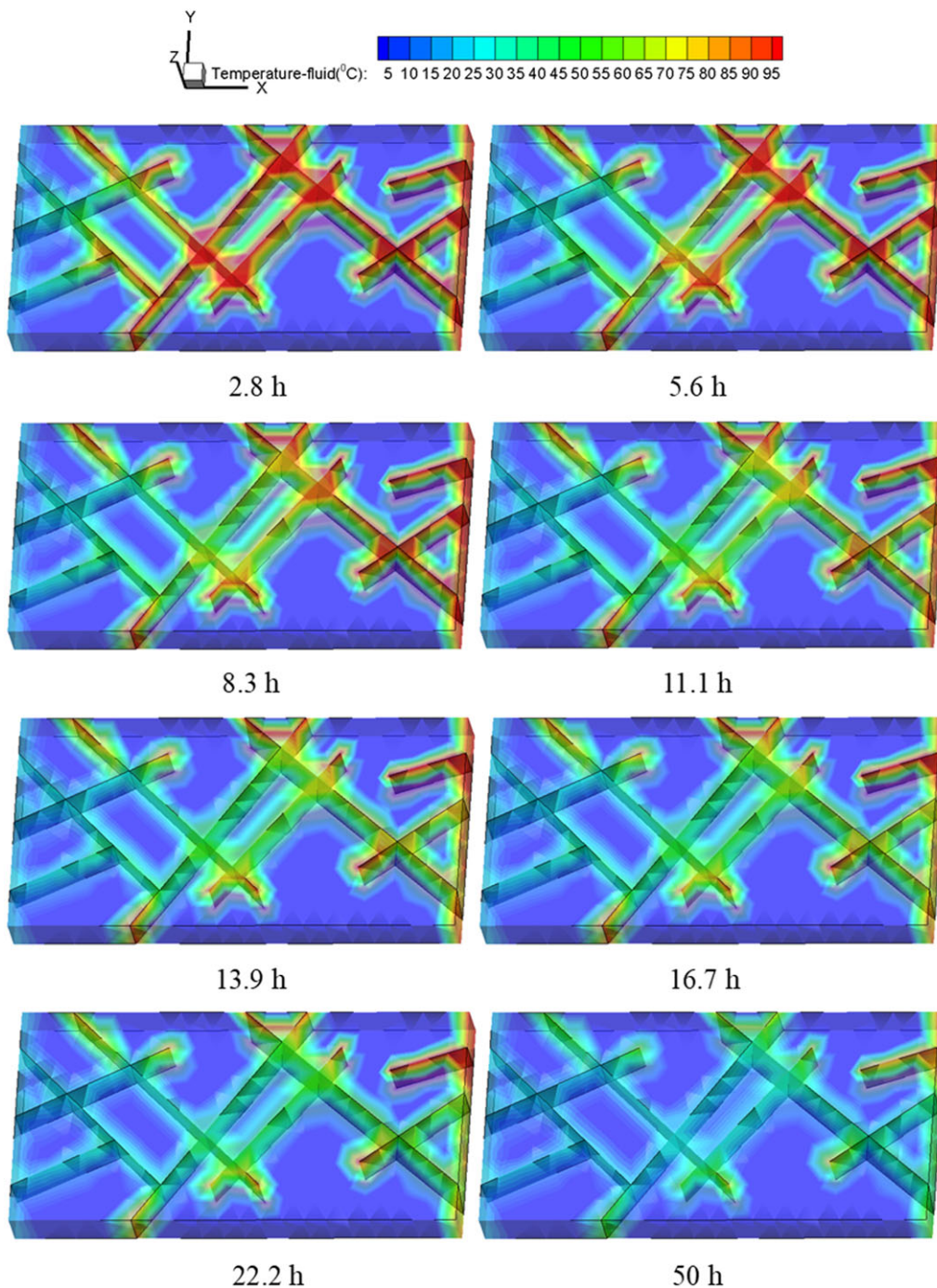


FIGURE 25 Temperature distributions in the fluid at different times [Colour figure can be viewed at wileyonlinelibrary.com]

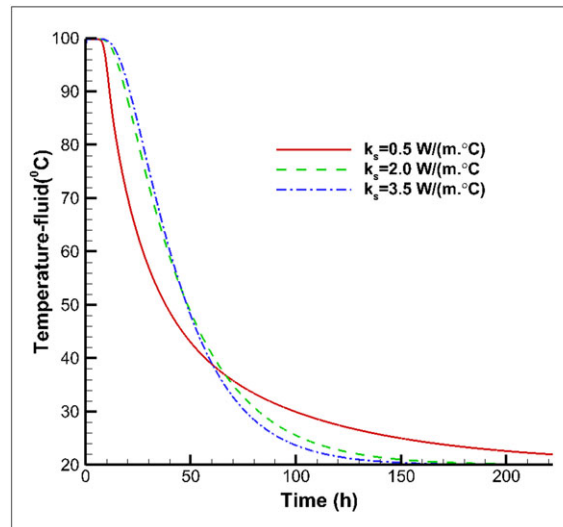


FIGURE 26 Effect of the thermal conductivity of the rock matrix on the fluid temperature at the outlet [Colour figure can be viewed at wileyonlinelibrary.com]

5.2.3 | Temperature field of the fluid in the fractures

As shown in Figure 25, the injected fluid is heated by the rock matrix, and the fluid temperature in the fracture increases gradually from left to right, as shown at 2.8 hours. With the continuous injection of cold fluid, the heat of the rock matrix is continuously carried away by the fluid. The temperature of the injected fluid decreases gradually as shown in Figure 25 at 50 hours, and the outlet temperature decreases gradually.

5.2.4 | Effect of the thermal conductivity of the rock matrix on the fluid temperature at the outlet

As shown in Figure 26, the greater the thermal conductivity of the rock matrix is, the later the thermal breakthrough occurs, and the less obvious the long tail effect is. The greater the thermal conductivity of the rock matrix is, the more slowly the outlet fluid temperature drops initially. This is because a high-temperature rock matrix with a high thermal conductivity heats the cold fluid rapidly. After a long period of time, the heat energy stored in the rock matrix is absorbed by the fluid, and the fluid temperature at the outlet decreases faster than that of the rock matrix with a low thermal conductivity.

6 | CONCLUSION

In this paper, a three-dimensional FDEM-TH model is established to simulate the coupled hydrothermal process in fractured rock. Four examples with analytical solutions are provided to verify the accuracy of the 3D coupled hydrothermal model. The numerical results agree well with the analytical results, verifying the model's accuracy. A simulation of a heat transfer experiment between fluid and rock is then presented. The numerical simulations also agree well with the experimental results. In addition, the effects of the fracture aperture, fluid viscosity, and pressure difference on the heat exchange between the fluid and the rock are investigated using the 3D coupled hydrothermal model. The results show that increasing the fracture aperture and pressure difference between the inlet and outlet promotes the heat exchange between the fluid and the rock, whereas increasing the fluid viscosity inhibits the heat exchange. Finally, an application of heat transport and fluid flow in rock containing a fracture network is presented to demonstrate the ability of this 3D model to perform coupled hydrothermal simulations. Considering the outstanding advantages of the FDEM in simulating rock fracturing, we can combine the 3D coupled hydrothermal model with the 3D FDEM to develop a three-dimensional analysis approach of rock fracturing driven by coupled THM effects.

ACKNOWLEDGEMENTS

This work was supported by the National Natural Science Foundation of China under grant numbers 11872340 and 11602006; the Beijing Natural Science Foundation under grant number 1174012; the Fundamental Research Funds for the Central Universities, China University of Geosciences (Wuhan) (CUG170657); the Chaoyang District Postdoctoral Science Foundation funded project under grant number 2016ZZ-01-08; and the National Natural Science Foundation of China under grant number 41731284.

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How to cite this article: Yan C, Jiao Y-Y. FDEM-TH3D: A three-dimensional coupled hydrothermal model for fractured rock. *Int J Numer Anal Methods Geomech.* 2018;1-26. <https://doi.org/10.1002/nag.2869>